ENGIN-X In-situ LCF Experiment
Pre-run Setup and Launch Procedure

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Operating guide based on beamline setup notes

Use this procedure as a practical setup checklist for the Instron / ENGIN-X in-situ fatigue experiment. Any beamline-specific interlocks, safety limits, or control-mode conventions should always be confirmed with the beamline scientist before execution.

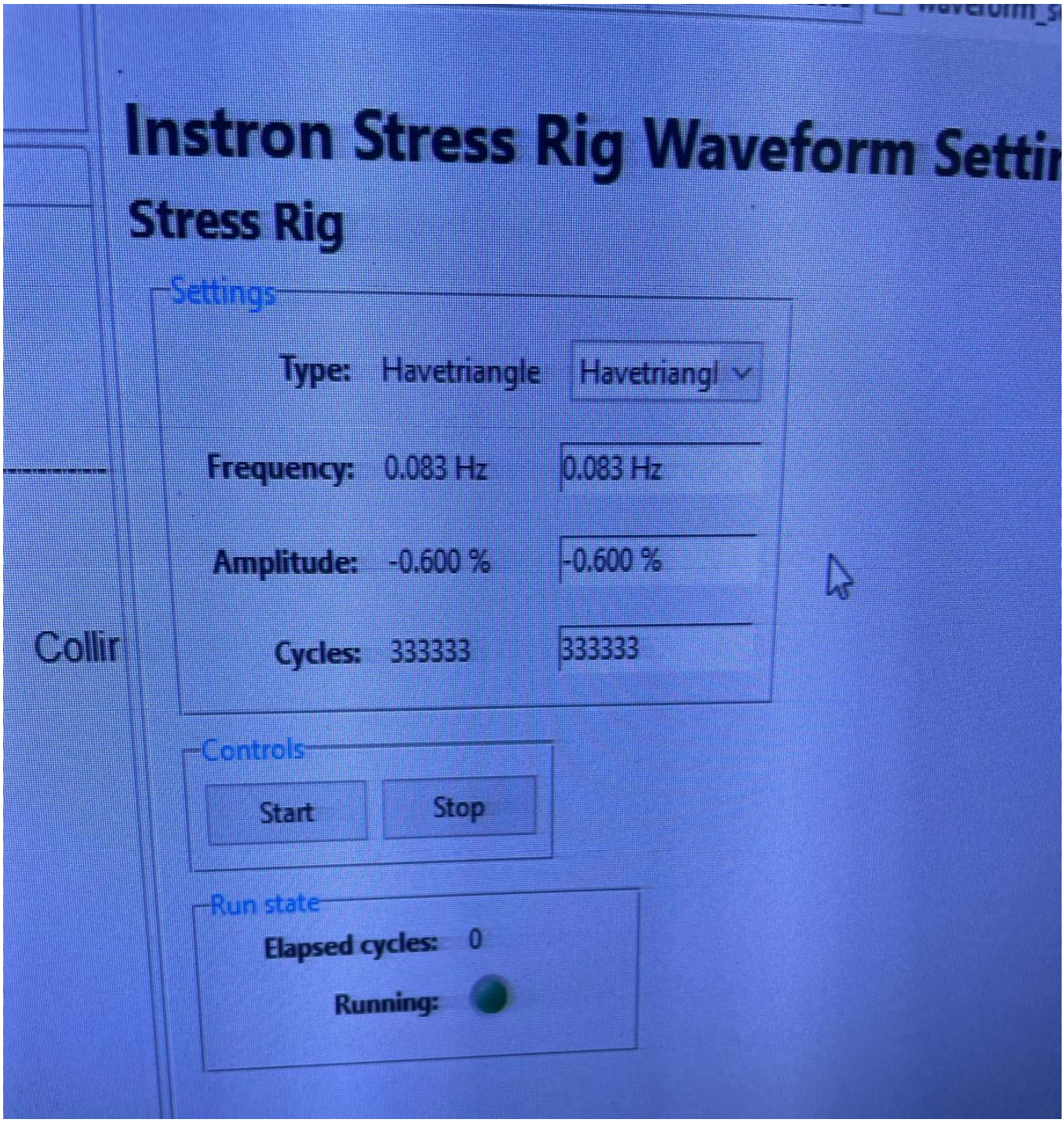
1. Before starting

- Confirm the correct ENGIN-X / Instron control computer is being used and that only one machine is communicating with the rig.
- Check that the correct sample ID, specimen dimensions, waveform settings, temperature target, and neutron measurement script are ready.
- Have the thermocouple, extensometer, grips, tools, and any sample protection hardware prepared before mounting the specimen.

2. Step-by-step procedure

1. **Zero the load with no sample installed**
   With no specimen mounted, use the Instron computer to balance **the load to zero in the setpoint window**. This gives a clean starting point before the specimen is installed.
2. **Mount the specimen and attach the thermocouple**
   Install the threaded sample in the rig carefully (with one side threaded all the way through), twist the other end grip to fit the threaded sample, and tighten the threaded sample with a set of M12 nuts to lock. Then, attach the thermocouple in the middle of the gauge length. Check that the sample is seated correctly and aligned before tightening fully to avoid buckling.
3. **Return to zero load in load control**
   On the Instron computer, work in load control and move the rig to zero load after the specimen has been mounted. This establishes the reference state before attaching the extensometer.
4. **Switch to position control (Engin-X GUI inside) and attach the extensometer**
   At zero load, change to displacement control on the inside ENgin-X GUI and then attach the extensometer carefully. Use the GUI readback to confirm the signal is stable before proceeding.
5. **Balance the extensometer on INSTRON software outside**
   Balance button in the strain-1 channel window (above calibration wizard).
6. **Return to load control and begin heating**
   Switch back to load control before heating. Start the heating stage and bring the specimen to the required test temperature.
7. **Soak at temperature and check strain offset (check)**
   After the specimen has soaked at temperature, inspect the strain reading again. If thermal drift has introduced an offset, first record the value. Only rebalance the strain channel if that is the agreed beamline practice for the experiment.
8. **Set strain limits**
   Enter the strain limits for the test, for example ±0.6% total strain range if that is the approved test condition.
9. **Set specimen geometry and cross-sectional area**
   Enter the specimen dimensions on both Instron computers. In the load → calibration/specimen geometry section, update the diameter so that the software calculates the correct cross-sectional area. Confirm that both the inside and outside control interfaces show the correct value.

10. **Configure the waveform on the inside Engin-X GUI computer**
    Using the ENGIN-X waveform setup screen (Synoptic → stress_rig.opi → waveform → stress rig setting), enter the required waveform parameters for the experiment. Verify control mode, waveform type, amplitude, rate or frequency as applicable, mean value, and cycle count settings before starting. Amplitude always in negative of the whole strain range (Peak-to-peak)



11. **Check the live readbacks before launch**
    Confirm that load, strain, displacement, temperature, and control mode look sensible. Make sure the rig is not left in an unexpected default mode such as position control after a previous trip or restart.
12. **Check specimen protection settings**
    Confirm whether specimen protection should be disabled for this run, and verify that the setting matches the beamline plan. Do not proceed until this has been explicitly checked.

13. **Launch the experiment script**
    Once the mechanical setup, thermal condition, waveform configuration, limits, and protection settings are all confirmed, launch the IBEX / Genie script for the experiment and monitor the first actions carefully.
14. **Observe the start of the run**
    Watch the first part of the experiment closely to confirm the rig enters the expected control mode, the waveform behaves correctly, and the neutron measurement sequence starts as intended.

3. Points to record in the logbook

- Sample ID, heat treatment condition, and specimen dimensions.
- Thermocouple attachment method and location.
- Zero-load balance status before mounting.
- Initial strain balance value and any post-soak strain offset.
- Test temperature and soak duration.
- Waveform settings used for the run.
- Whether specimen protection was on or off.
- Time the script was launched and any abnormalities at startup.

4. Caution / confirmation items

- If the strain channel is rebalanced after thermal soak, record the offset first so the action is traceable.
- If two control computers can access the rig, ensure only the intended system is communicating during setup and execution.
- If the waveform display or units look inconsistent, confirm the actual rig behaviour from the live readbacks rather than relying on a single display field.
- If the rig has tripped or been powered down, confirm the control mode and interlocks before restarting the test.